

Safe System of Work

Managing Dust in Construction

Ref. SSoW-051	Author.	Lee Hardy	Date:	01.09.2022
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Record of Amendments			

Next Review Due Date:	31.08.2026
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Overview

The following information provides guidance on the company's Safe System of Work for works that involve the management of dust in construction.

All employees have a responsibility for their safety and the safety of others by adhering to these safety systems.

The Operative must always refer to any associated Risk Assessment and Method Statements prior to starting works.

Hazards Identified

There are 3 main types of construction dust:

1. Silica Dust

Silica is a natural mineral present in large amounts in things like sand, sandstone and granite. It is also commonly found in many construction materials such as concrete and mortar. The silica is broken into very fine dust (also known as Respirable Crystalline Silica or RCS) during many common tasks such as cutting, drilling and grinding. It is often called silica dust

2. Non-silica dust

There are a number of construction products where silica is either not found or present in very low amounts. The most common ones include gypsum, cement, limestone, marble and dolomite. This dust is also mixed with silica dust when cutting things like bricks.

3. Wood dust

Wood is widely used in construction and is found in two main forms: softwood and hardwood. Wood-based products are also commonly used including MDF and chipboard

Anyone who breathes in these dusts should know the damage they can do to the lungs and airways. The main dust related diseases affecting construction workers are:

- **Lung cancer**
- **Silicosis**
- **Chronic Obstructive Pulmonary Disorder (COPD)**
- **Asthma**

Personal Protective Equipment (PPE) Requirements



There are a number of things that can happen when using water or vacuum extraction that can stop the dust being properly controlled. Even if this does not happen, some tasks are so dusty that enough escapes into the air to still be a risk. For this reason, some form of respiratory protective equipment (RPE), usually in the form of a mask, is also needed for high-risk tasks such as:

- using a cut-off saw, grinder or wall chaser on material containing silica
- using powered cross-cut saws and sanders on hardwood, red cedar or MDF
- sanding softwood in an enclosed space

Masks are available that provide different levels of protection. There are two main performance types you should ask your supplier/ employer for if working with construction dusts: FFP2 and FFP3. FFP3 is the most advisable type to use if you are doing work that does or could create high dust levels or involves silica or wood dust (the more hazardous substances).

How Much Dust Can Harm Me?

To protect your lungs the Control of Substances Hazardous to Health (COSHH) sets a limit on the amount of these dusts that you can breathe. These limits are not a large amount of dust. To give an example the image below shows the maximum amount of silica you can breathe when averaged over a normal working day.

When compared to a penny it is tiny – like a small pinch of salt:



This limit is the legal maximum, the most you can breathe **after** the right controls have been used. For tasks that can create high levels of silica and wood dust these controls have to be very good as the risks from these dusts are high

People often think that construction dust is not a problem if they are working outside because it will just blow away. While the wind will have some effect on the level of dust many tasks involve the person working close to the part of a tool where the dust is being made. With very dusty tasks this means that they can still breathe a lot of harmful dust.

The risk of lung disease is linked to people who regularly breathe construction dust over a period of time, not on the odd occasion. However, breathing construction dust even over a short period of time is not nice (for example when passing some street works) and could cause a reaction in someone who already has asthma or another existing breathing problem. It therefore should be avoided by using the right controls and making sure that members of the public are not close to the work when it is going on.

Controlling Construction Dust

You should look at ways of limiting the amount of dust you could make before you start work. For example, you could:

- use the right size of building materials, so less cutting or preparation is needed
- use a less powerful tool
- using a different method of work altogether
- Reduce speed limits on site

Even if you can stop some of the dust this way you may need to do other work that could still produce high amounts of dust. In these cases, the most important thing is to stop the dust getting into the air. There are two main ways of doing this which both give very good results:

- **water** – water damps down dust clouds. However, it needs to be used correctly. This means enough water for the whole time that the work is being done. Just wetting an area of ground before cutting does not work., but it will reduce dust clouds caused by moving wagons or plant machinery.
- **vacuum extraction** – specially designed tools can be fitted with an industrial vacuum unit that sucks the dust away as it is being created and stores it until emptied.

Wagons and plant machinery should maintain a slower speed when driving in construction sites and quarries, especially on hot days when dust will be more prevalent.

Dust suppression on construction sites needs to be by way of dampening down – either by pressure washer, road sweeper or towed water bowser.

